

SECTION X

OVERHAUL INSTRUCTIONS

10-1. PURPOSE.

10-2. The purpose of this section is to provide instructions for overhauling the retrieve engine and the torque converter to restore the equipment to a safe and efficient operational status.

10-3. GENERAL INSTRUCTIONS.

10-4. PREOVERHAUL. Refer to NAV-AIR 51-5-31, PART A, for the preoverhaul conference, preliminary inspection, test of equipment, and the overhaul report that apply to the E-28 Emergency Runway Arresting Gear.

10-5. OVERHAUL PROCEDURE. The following procedural listing pertains to the retrieve engine and the torque converter during overhaul:

a. Clean all external parts and electrical equipment with solvent, Specification P-D-680, Type II. Dry with external parts with a clean, lint-free cloth or blow dry with low-pressure compressed air.

b. Discard and replace the following parts whenever encountered:

1. All O-rings and gaskets.
2. All cotter pins.
3. All lockwire.

c. Lubricate the retrieve engine and torque converter according to the lubrication chart (figure 8-1) in addition to any lubrication specified during component assembly.

d. Torque tables (tables 10-1 thru 10-3) are provided for tightening various components on the retrieve engine and the torque converter to the proper values.

10-6. RETRIEVE ENGINE OVERHAUL.

10-7. These procedures cover the removal and reassembly of the engine components. The components are listed in the order of engine disassembly and are assembled and reinstalled in reverse order unless otherwise noted. Crankcase oil should be drained from engine before overhaul.

10-8. ACCESSORIES. All accessories should be removed from the engine and disconnections made first as follows:

- a. Remove air cleaner, oil filter, magneto, starter, and generator.
- b. Disconnect fuel hose from engine and fuel tank.
- c. Disconnect all control lines and wiring leading to control panel.

NOTE

Clean and repair accessories before engine reassembly.

10-9. FLYWHEEL. The flywheel removal is accomplished as follows:

- a. After the flywheel screen has been removed, the flywheel nut and the drive pulley should be taken off so that the flywheel can be removed from the crankshaft.

TABLE 10-1

RETRIEVE ENGINE, TORQUE TABLE

<u>Part</u>	<u>Torque Value lb. ft.</u>
Spark plugs	25 to 30
Manifold nuts	40 to 50
Cylinder head screws	25 to 32
Gear cover capscrews	14 to 18
Oil pan mounting screws	6 to 9
Connecting rod nuts	28 to 32
Cylinder block mounting nuts	62 to 78
Main bearing plate capscrews	25 to 30

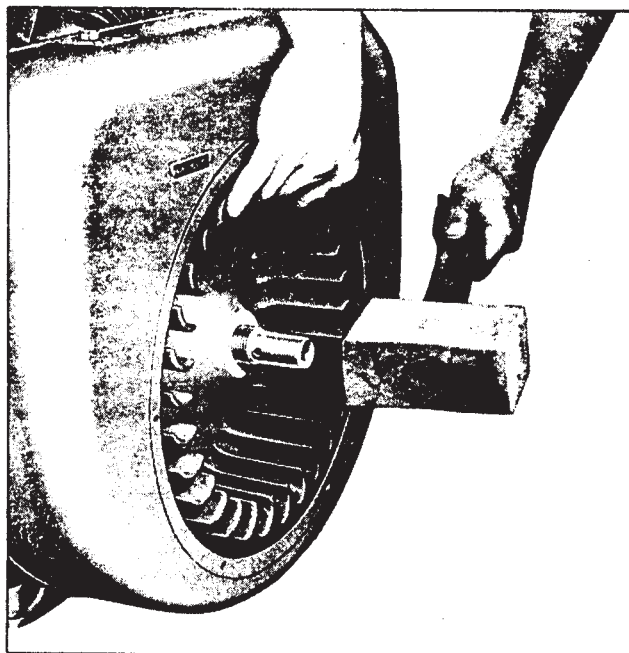


Figure 10-1. Flywheel Removal

b. Take a firm hold on the flywheel fins, pull outward and at the same time strike the end of the crankshaft with an aluminum hammer (see figure 10-1). The flywheel will slide off the taper of the crankshaft. Do not use a hard hammer as it may ruin the crankshaft and bearings.

CAUTION

When reassembling the flywheel, be sure the Woodruff key is in position on the shaft and that the keyway in the flywheel is lined up accurately with the key.

10-10. AIR SHROUDING. To disassemble air shrouding, refer to figure 10-2. First remove cylinder head covers and the screws mounting the flywheel shroud to the lower cylinder shrouds and cylinder heat

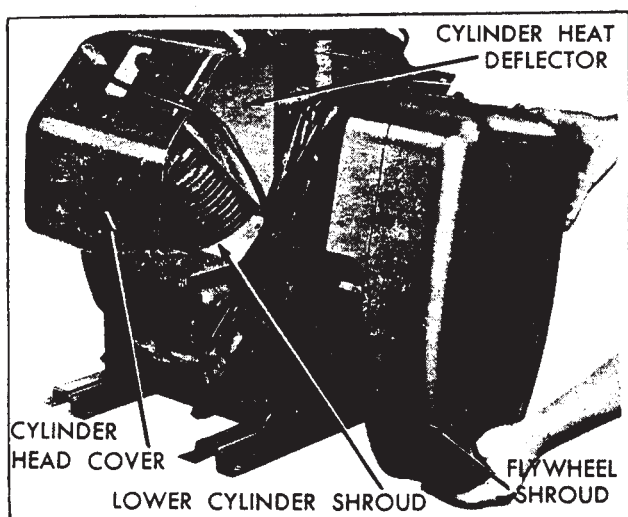


Figure 10-2. Flywheel Shroud Removal

deflectors; then remove the screws holding the flywheel shroud to gear cover.

10-11. CARBURETOR AND MANIFOLDS.

The carburetor and manifold can be removed as a complete unit as shown in figure 10-3. On engine reassembly, torque the mounting nuts as specified in table 10-1. Overtightening may cause flanges to break.

NOTE

Engine can be lifted for removal by inserting hook under exhaust manifold web before the carburetor and manifold unit is disconnected. Engine must be unbolted from torque converter at clutch housing and shoved forward before lifting with a hoist.

10-12. CYLINDER HEAD. The cylinder head must be removed if it is necessary to regrind valves, or to do work on the piston, rings or connecting rod.

a. All of the cylinder head screws are plainly in view and can be easily removed. Screws of different lengths are used but these can be properly reassembled according to the various lengths of cylinder head bosses.

b. Before reassembling the cylinder head, all carbon and lead deposits should be removed.

c. Use a new cylinder head gasket on reassembly, as the old gasket will be compressed and hard so that it may not seal properly.

d. Use a mixture of graphite and oil on the cylinder head screws, to prevent them from rusting tight against the cylinder block.

e. Torque cylinder head screws on reassembly as specified in table 10-1.

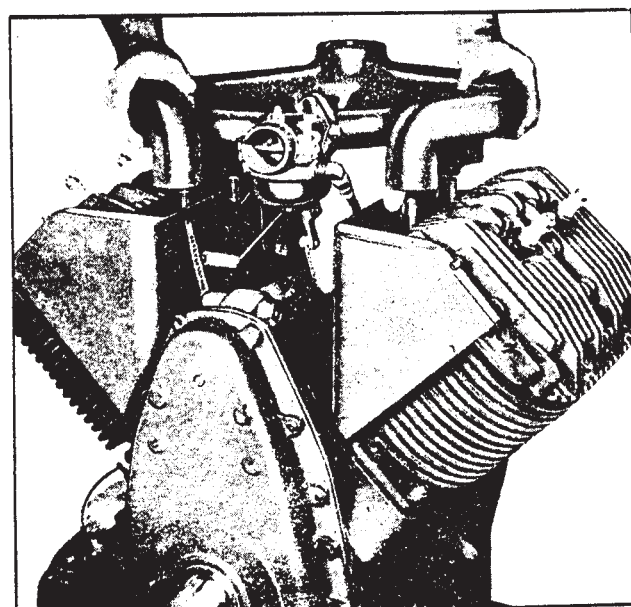


Figure 10-3. Carburetor and Manifold Removal

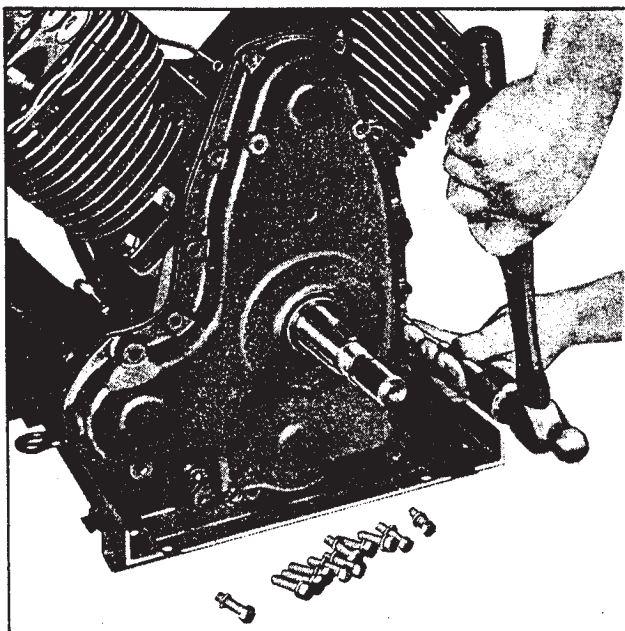


Figure 10-4. Gear Cover Dowel Pin Removal

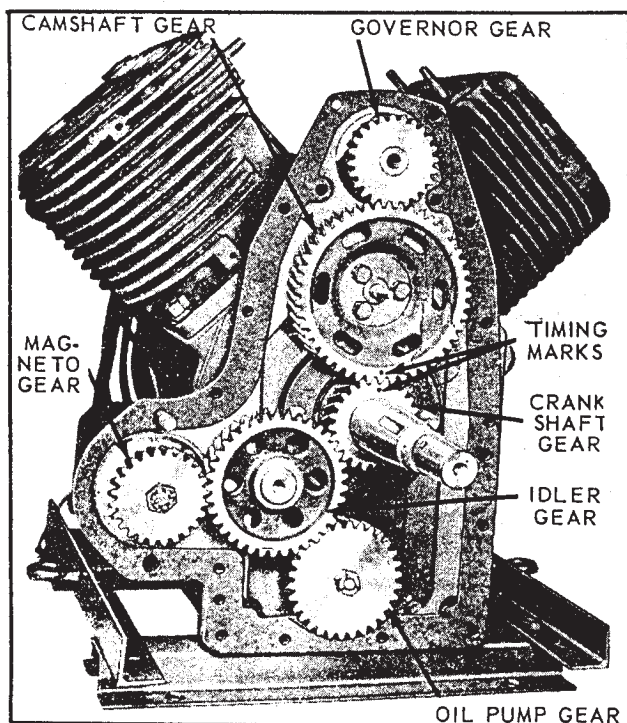


Figure 10-5. Gear Train

NOTE

After complete engine assembly and engine run-in, retorque the head screws.

10-13. GEAR COVER. Disconnect the governor linkage and remove the governor assembly. Remove gear cover screws and drive out the two dowel pins as shown in figure 10-4. The gear cover can then be taken off, exposing the timing gears as shown in figure 10-5. On reassembly, torque gear cover capscrews as specified in table 10-1.

10-14. GEAR TRAIN. With the removal of gear cover and oil sling, the gear train will be exposed as shown in figure 10-5. Remove camshaft thrust plunger and spring, to prevent their being lost.

NOTE

Future reference can be made to figure 10-5 when assembling crankshaft and camshaft, as accurate location of the timing marks is essential.

10-15. IDLER GEAR AND SHAFT. Remove the Allen-head setscrew, on the magneto side of the crankcase, which locks the idler shaft in position. With the use of a gear puller, the idler shaft and idler gear assembly can be removed from the crankcase. (See figure 10-6.)

NOTE

In reassembly, be sure oil groove in shaft is facing up. Drive shaft into crankcase with soft metal hammer and maintain a .003 to .004 inch clearance between idler gear and shoulder of shaft.

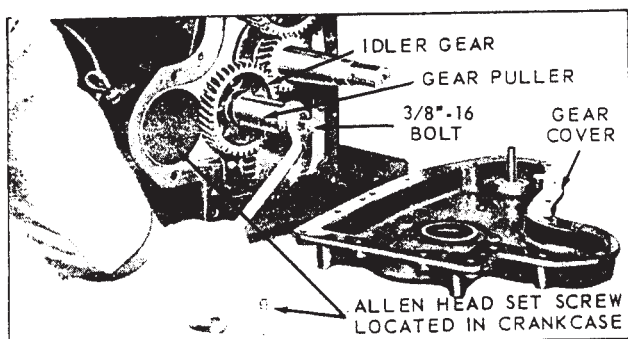


Figure 10-6. Idler Gear Removal

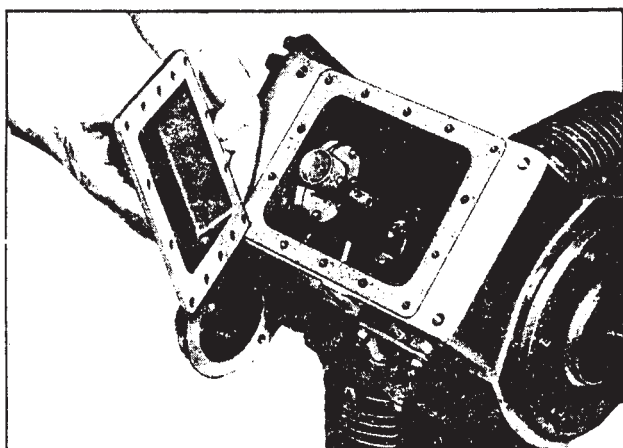


Figure 10-7. Oil Pan Removal

10-16. OIL PAN. The engine can now be inverted so that the supports and oil pan can be removed (figure 10-7). On re-assembly, mount deep end of oil pan towards oil pump and refer to table 10-1 for mounting-screw torque value.

10-17. OIL PUMP. Remove the oil pump as follows:

a. Take out the slotted pipe plug, and then with a 5/32 inch Allen wrench, remove the oil pump lockscrew, as shown in figure 10-8.

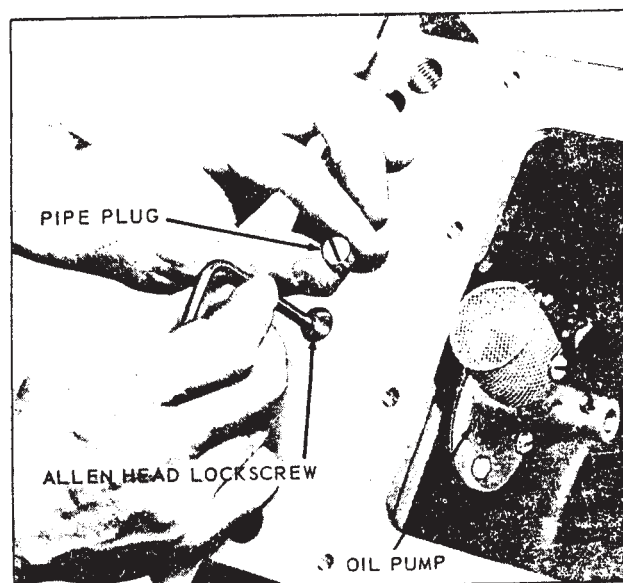


Figure 10-8. Oil Pump Lockscrew and Pipe Plug Removal

b. Remove locknut holding oil pump driving gear to shaft.

c. With a soft brass rod or punch, drive shaft through gear as shown in figure 10-9. The oil pump can then be withdrawn toward center of crankcase.

10-18. PISTONS AND CONNECTING RODS. The connecting rod bolts are accessible after the oil pump has been removed. To disassemble and reassemble the pistons and connecting rods, proceed as follows:

a. Remove the Palnuts and hexagon nuts; then by tapping the ends of the bolts lightly, being careful not to mar the threads, the connecting rod cap can be freed from the bolts.

b. The rod with the piston can now be pushed up through the cylinder. Be careful not to score the crankshaft journals, by allowing the rod bolts to strike or

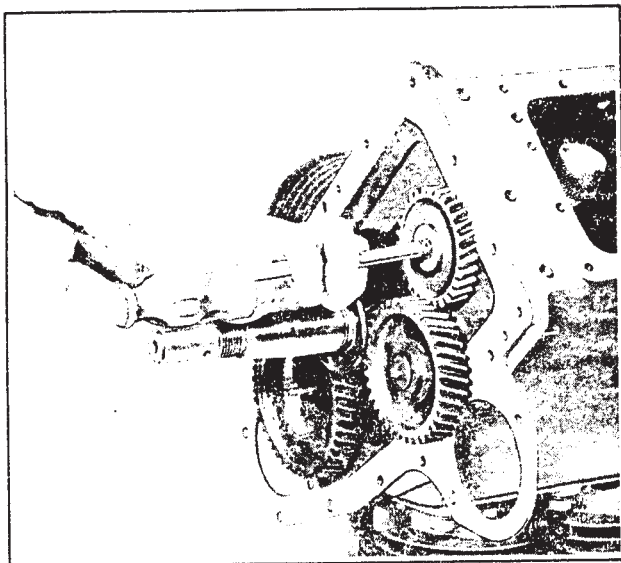


Figure 10-9. Oil Pump Drive Gear Removal

scrape across them, when removing the connecting rod and piston assemblies.

c. Replace the caps on the rods immediately so that they are in the correct position for reassembly. A number is stamped on the side of the rod and cap to match each connecting rod with its corresponding cap. These numbers must be on identical sides of the rod in reassembly. (See figure 10-10.)

d. The connecting rods have removable shell bearings. New shell bearings must be used in reassembly. Exercise care in replacing bearing that both bearing halves are in place.

e. The piston skirt is cam-ground to an elliptical contour. Clearance between the piston and cylinder must be measured at the center of the thrust face of the piston skirt. Refer to figure 10-13 for proper clearance. The thrust faces on the piston skirt are 90° from the axis of the piston pin hole. (See figure 10-11.)

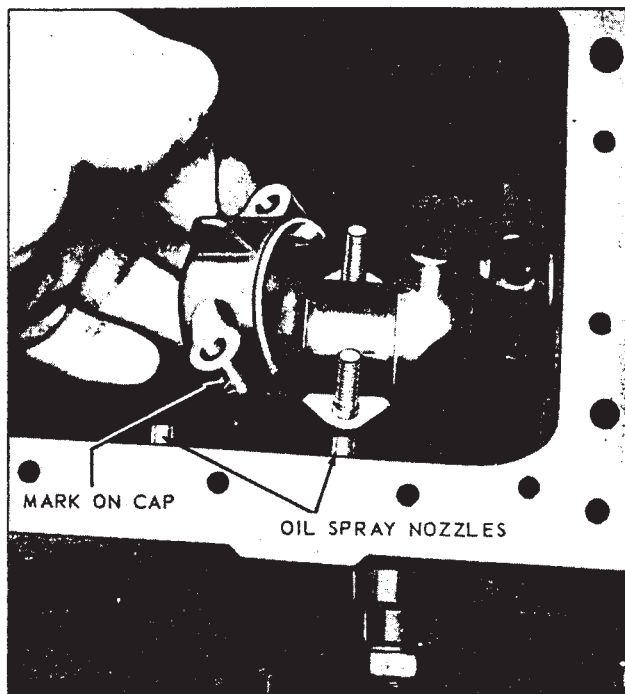


Figure 10-10. Connecting Rod Cap Removal

f. When reassembling the piston and connecting rod to the engine, be sure the arrow on the top of the piston is pointing in the direction of crankshaft rotation

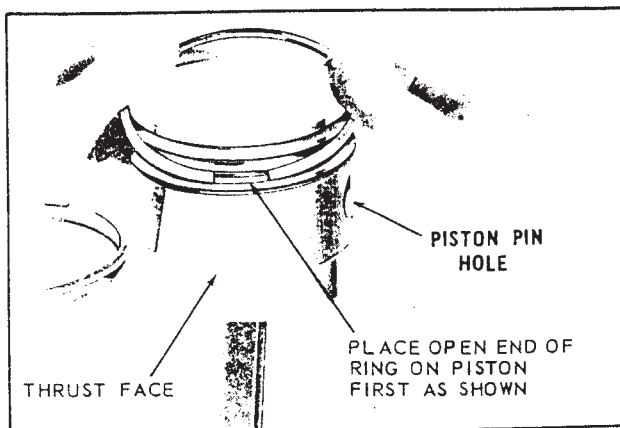


Figure 10-11. Piston Ring Installation

(clockwise when viewing the flywheel end of the engine).

g. Tighten connecting rod nuts according to torque value in table 10-1, then install Palnuts and tighten with wrench 1/4-turn past finger-tight position.

CAUTION

Be sure piston and connecting rod assemblies are put back into the same bore from which they were removed.

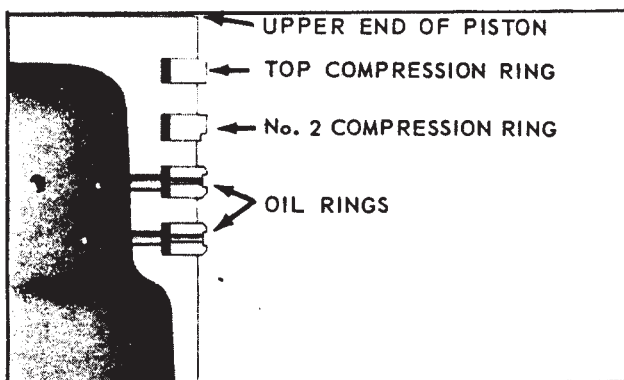


Figure 10-12. Piston Ring Identification

10-19. PISTON RINGS. Install new piston rings as follows:

a. Install rings by placing the open end of the ring on piston first, as shown in figure 10-11. Spread ring only far enough to slip over piston and into correct groove, being careful not to distort ring.

b. The Number 2 compression ring must be installed on the piston with the scraper edge down, otherwise oil pumping and excessive oil consumption will result. Refer to figure 10-12.

PISTON TO CYLINDER AT PISTON SKIRT THRUST FACES		.0052 to .0062"
PISTON RING COMPRESSED GAP		.025 to .035"
PISTON RING SIDE CLEARANCE IN GROOVES	Top Ring	.002 to .004"
	2nd Ring	.0015 to .0035"
	3rd, 4th Groove Oil Rings	.001 to .003"
CONNECTING ROD TO CRANK PIN - SIDE CLEARANCE		.008 to .015"
CONNECTING ROD SHELL BEARING TO CRANK PIN		.0015 to .003"
PISTON PIN - TO PISTON TO CONNECTING ROD BUSHING		.0000 to .0003" .0002 to .0007"

2.1230
2.1225
DIA. GRIND
1.355
1.350 WIDTH
STANDARD CRANK PIN DIMENSIONS

Figure 10-13. Piston, Ring and Rod Clearance Chart

c. Use a suitable ring compressor in reassembly and stagger the piston ring gaps 90° apart around the piston. Oil the pistons, rings, wrist pin, rod bearings and cylinder walls before assembly.

10-20. CYLINDERS. The cylinder blocks can now be removed from the crankcase if necessary. In reassembling, put the blocks back on the same side from which they were removed. Clean all dirt and other deposits from fins before reassembling. If the cylinders are worn more than .005 inch over-size, they should be reground and fitted with oversize pistons and rings. Tighten cylinder block mounting nuts according to torque value in table 10-1.

10-21. VALVES. The valves are disassembled and ground as follows:

a. Use the following procedure to remove the valves:

1. Remove the valve tappet inspection plate and compress the valve springs with a standard automotive type valve lifter, as shown in figure 10-14. If cylinder block is still attached to the engine, insert a rag in the opening at the bottom of the valve chamber so the roto-cap and valve spring seat retaining locks do not fall into the crankcase.

2. Remove the roto-caps, valve spring seat retaining locks, seats, springs, valves and clean these, as well as the ports and guides, of all carbon and gum deposits.

3. Tag each valve so that in reassembly they will be mounted in the same guide they were removed from.

NOTE

Nonpositive type roto-caps are provided on the ends of the valve stems. In reassembly, be sure to assemble the roto-caps to the end of the valve stems.

b. Grind and lap the valves as follows:

1. The valve face is ground at 45° to the vertical center line of the valve stem and the valve seat insert should also be ground at a 45° angle.

2. After grinding, valves and inserts should be lapped with a suitable lapping compound or they will leak due to improper seating within the first few hours of operation. After valve seats have been cleaned, apply lapping compound to the valve face and put the valves back into their guides.

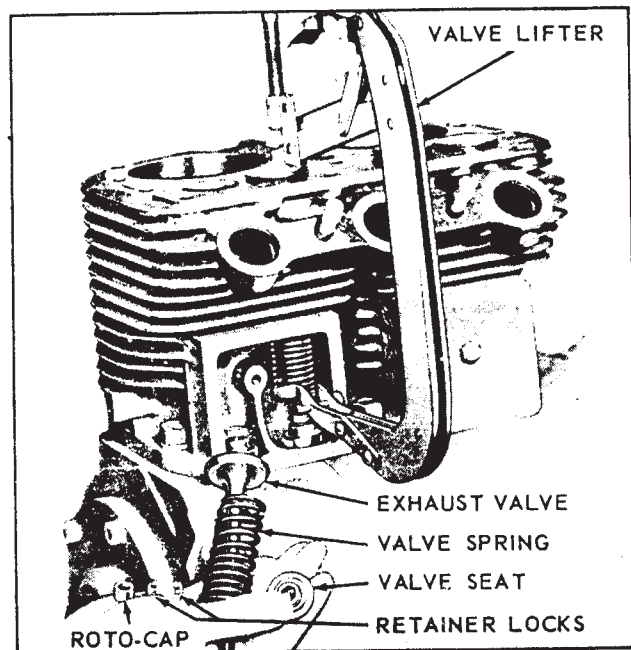


Figure 10-14. Valve Spring Compression

3. Lap the valves by rotating them back and forth with a reciprocating advancing valve tool. Occasionally lift the valves and reseat them in a different position to insure a uniform seat which will show entirely around the valves.

4. After valves have been lapped in evenly, remove them from the block and wash the valves and block thoroughly with gasoline or kerosene.

c. The cylinder blocks have replaceable valve guides. The valve stems have a clearance of .0025 to .0045 inch in the guides. When the clearance becomes .008 inch, the guides should be driven out and replaced with new guides.

NOTE

The valves and valve seat inserts are of Stellite material.

10-22. **CRANKSHAFT.** The crankshaft is removed and reassembled as follows:

a. To remove the crankshaft, first remove the six capscrews in the main bearing plate at the take-off end of the engine. The plate can then be pried off and the crankshaft removed from that end of crankcase. (See figure 10-15.)

CAUTION

Be sure to keep shims and gaskets in place as these are necessary to give the proper end play to the tapered roller bearings on the crankshaft. This end play should be .002 to .004 inch when engine is cold. There is practically no wear in these bearings so that no readjustment is necessary after proper assembly.

b. When reassembling crankshaft, the timing marks on the crankshaft gear and camshaft gear must match up (refer to figure 10-5); otherwise, engine will not operate properly, or if timing is off considerably, engine will not run at all.

c. Mount main bearing plate in the correct position in reassembly. The word 'TOP' is cast on the outside of the plate, and should be mounted in this position. Mounting the main bearing plate upside down would prevent the main bearing from being properly lubricated. Tighten main bearing plate screws to torque value specified in table 10-1.

10-23. **CAMSHAFT.** Remove and reassemble the camshaft as follows:

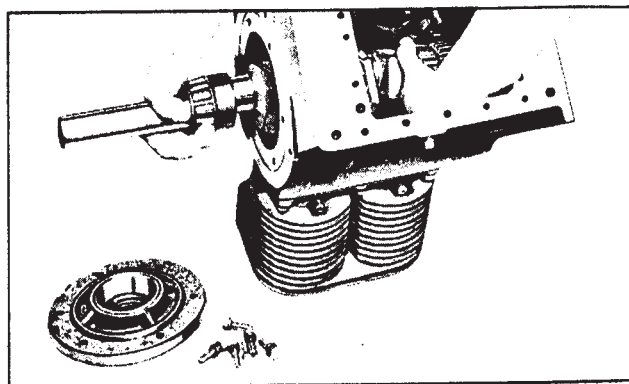


Figure 10-15. Crankshaft Removal

a. Remove all valve tappets and withdraw camshaft from crankcase (figure 10-16).

b. Camshaft gear is removed by taking out the three screws and lockwashers, and then prying the gear off the end of the camshaft.

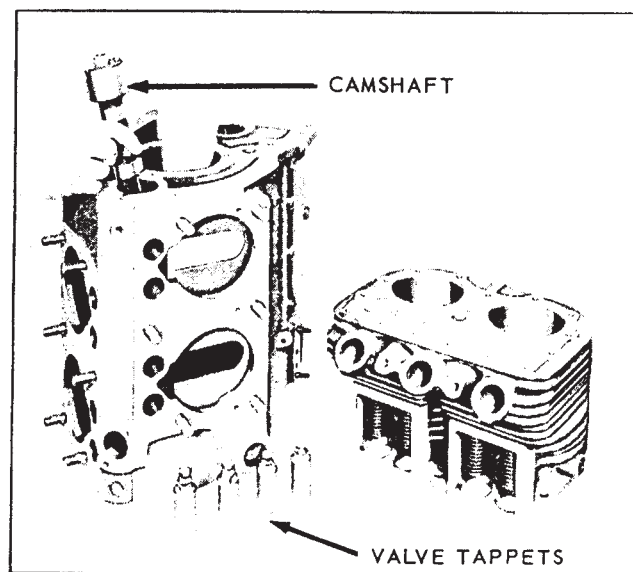


Figure 10-16. Camshaft Removal

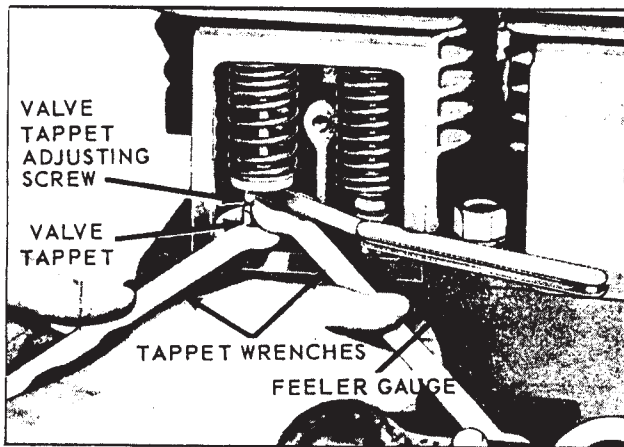


Figure 10-17. Valve Tappet Adjustment

NOTE

The camgear has offset mounting holes to provide accurate assembly for valve timing. The gear can only be put on the correct way for matching up the timing mark with that of the crankshaft (see figure 10-5).

c. On reassembly, be sure the spring and plunger are in place in the end of the camshaft, as these hold the camshaft in position endwise.

10-24. VALVE TAPPET ADJUSTMENT.

The valve tappets are adjusted after the cylinder block assemblies have been mounted to the crankcase. (See figure 10-17.) With the tappets at their lowest positions and the engine cold, clearance should be .008 inch on the inlet valves and .016 inch on the exhaust valves.

10-25. FUEL PUMP. (Refer to figure 13-15). The fuel pump efficiency can be restored by using a fuel pump repair kit for rebuilding. This is accomplished as follows:

a. Remove the fuel lines and the two mounting bolts which hold the pump to the engine. Take the pump to a work bench or suitable place.

b. Disassemble fuel head on pump and rebuild as follows:

1. With file make an indicating mark across a point at the union of castings (7 and 23). This is a positive location of the fuel line positions when re-assembling. Remove six assembly screws (6) and remove fuel head. Dome bolt (1) is taken out, and the dome (3), dome gasket (4) and filter screen (5) disassembled.

2. Turn head (7) over and remove three screws (13). Remove (8, 9, 10, 11, and 12) valve assemblies noting their positions. Discard items (8, 9 and 11.)

3. Clean head thoroughly with gasoline and a fine brush.

4. Holding fuel head (7) with diaphragm surface (containing 6 clearance holes) up, reassemble the valve spring retainer (10) into position, deepest hole. Place new valve plate gasket (11) in position around this hole. Stand on end one of the new valve springs (9) in the retainer. Place a new valve (8) on top of this spring.

5. Place the other new valve (8) on top of brass valve seat next to the previously assembled parts, stand the other new valve spring (9) on top of this valve.

6. Carefully place valve plate (12) over these assemblies so the large raised ring is nearest the valve gasket (11). Insert and carefully tighten the three assembly screws (13).

7. Carefully clean and replace the filter screen (5).

8. Place new dome gasket (4) into the dome (3) and put this unit into position on the fuel head (7).

9. Put new dome bolt gasket (2) on the dome bolt (1), insert in the hole in the dome (3) and tighten properly.

10. Set fuel head assembly aside in a clean place.

c. Rebuild the lower section of fuel pump as follows:

1. Using a screwdriver inserted into the coils of rocker arm spring (17) remove this spring and discard.

2. Hold the mounting bracket (23) in the left hand with the rocker arm toward your body and the thumb nail on the end of the link (19). With the heel of right hand on the diaphragm (14), compress the diaphragm spring (16) at the same time pulling toward your body. This action will unhook the diaphragm from the link (19) so it can be removed.

3. Clean the mounting bracket (23) with gasoline.

4. Place the remaining new cork gasket (15) on the corresponding surface of the mounting bracket (23). Assemble the new diaphragm operating spring (16) standing it into (23). Now repeat in reverse step 2 using the new diaphragm, rocker arm, pin, linkage and bushing. Assemble new rocker arm spring.

d. Install the fuel pump on engine as follows:

1. Mount this assembly back on the engine in the position from which it was removed, using the new mounting gasket which is the last piece of the repair kit.

2. Crank the engine over to a position where the diaphragm (14) is lying flat on the mounting bracket (23). Place the fuel head (7) back in position so the indicating marks of step b.1 are in line, and start the six assembly screws approximately three turns. Again crank the engine over to a position where the diaphragm (14) is pulled down into (23) mounting bracket and tighten the six assembly screws (6) tightly.

3. Connect the fuel lines.

10-26. GOVERNOR. The governor can be disassembled from the engine by first removing the governor housing, after which the entire governor can be withdrawn from the gear cover. The construction of the governor can be best seen from the sectional drawing of the engine (figure 1-5). The governor is adjusted as described in paragraph 8-20 after reinstallation on engine.

10-27. TORQUE CONVERTER OVERHAUL.

10-28. PRELIMINARY PROCEDURE. The oil should be drained from the torque converter before the overhaul. The oil drain plug (see figure 1-6 and 10-18) is located on the front and bottom of the converter housing.

10-29. INSPECTION TABLE. Table 10-4 lists the inspection procedures and action to be taken on component disassembly.

10-30. WEAR LIMITS. The wear limits points of measurement on the torque converter are shown in figure 10-30. For the spring wear limits, refer to table 10-5.

10-31. OVERHAUL EQUIPMENT. Proper equipment should be available before overhauling the torque converter. This should include a suitable hoist, appropriate

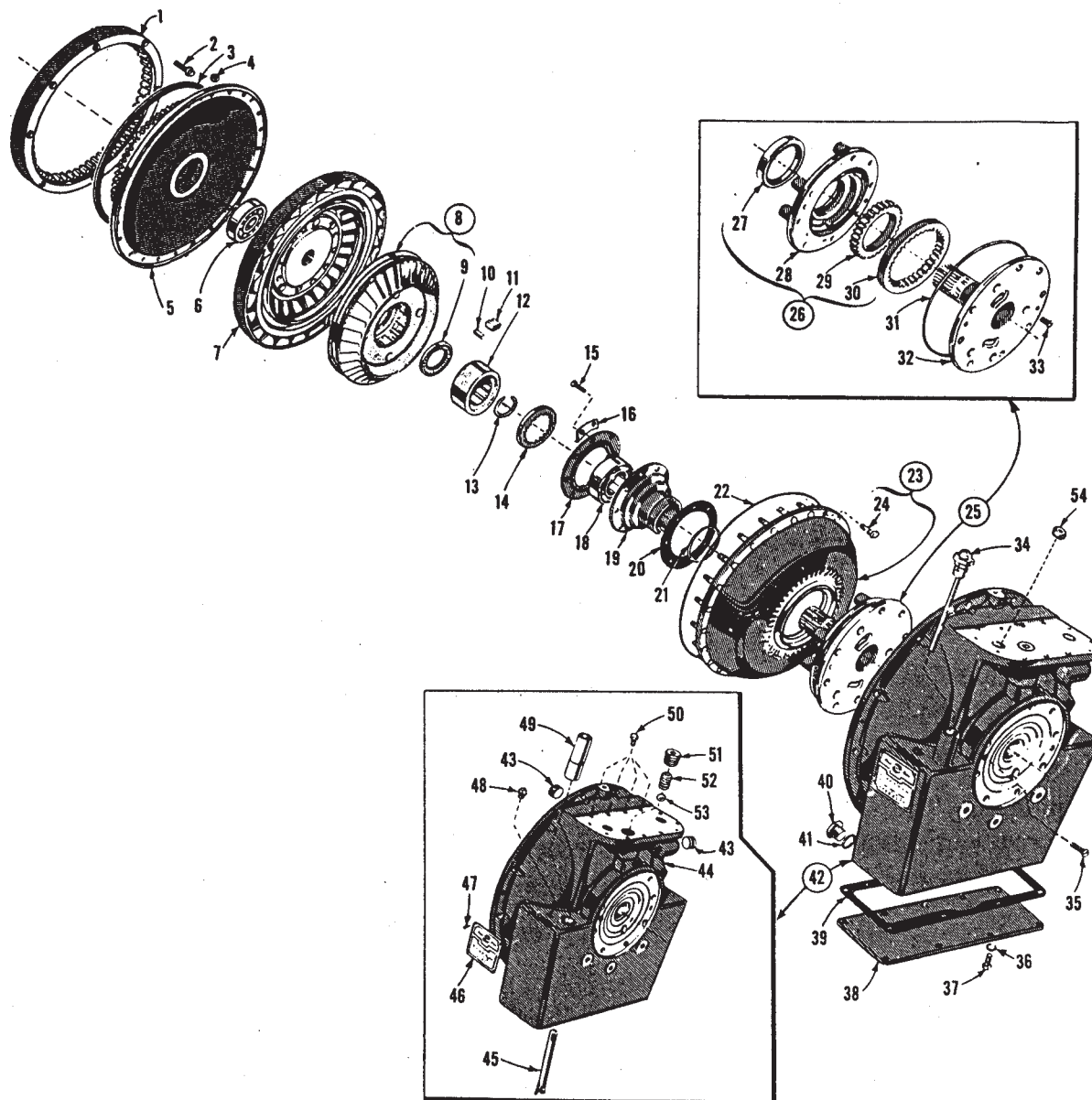


Figure 10-18. Torque Converter-Exploded View

hand tools, receptacles for small parts, and a press. A disassembly table (figure 10-31) and an overhaul stand (figure 10-32) are helpful but not essential.

10-32. OIL COOLER. (Refer to figure 10-19). The oil cooler is removed and re-installed as follows:

a. To remove cooler, use a 1/2-inch wrench and remove twelve bolts from the oil cooler cover (1). Remove the cover

(2) and gasket (3). Remove the core assembly and gasket (4 and 5).

b. To install the cooler, proceed as follows:

1. Install gasket (5) and cooler core (4) onto the mounting pad on top of the converter housing.

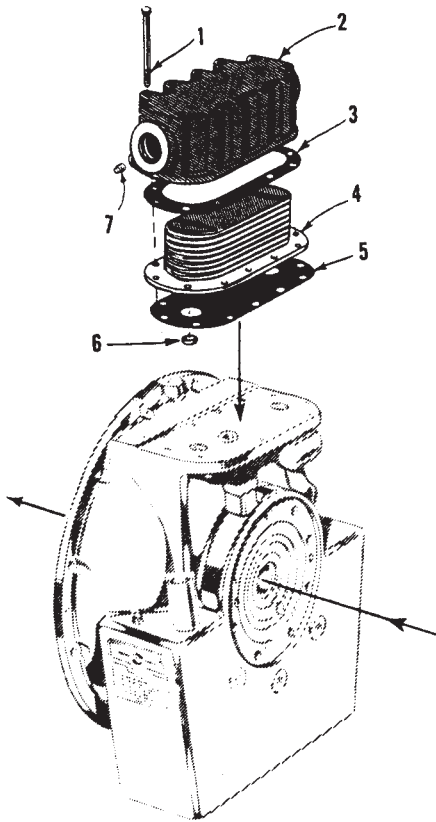
2. Install cover gasket (3) and cover (2) over the cooler core.

1 - Ring gear	20 - Gasket	38 - Sump plate
2 - Special bolt	21 - Hook-type seal ring	39 - Sump plate gasket
3 - Seal ring	22 - Seal ring	40 - Drain plug
4 - Lock nut	23 - Converter pump assembly	41 - Gasket
5 - Gear and cover	24 - Special bolt	42 - Converter housing assembly
6 - Bearing assembly	25 - Charging oil pump assembly	43 - Pipe plug
7 - Turbine assembly	26 - Gear and body assembly	44 - Converter housing
8 - Stator assembly	27 - Oil seal	45 - Oil cooler return tube
9 - Thrust bearing	28 - Oil pump body	46 - Name plate
10 - Stator freewheel roller	29 - Oil pump drive gear	47 - Screw
11 - Freewheel roller spring	30 - Oil pump driven gear	48 - Pipe plug
12 - Freewheel roller race	31 - Seal ring	49 - Oil filler sleeve
13 - Snap ring	32 - Stator support (ground sleeve)	50 - Bolt
14 - Roller race spacer	33 - Flat-head screw	51 - Plug
15 - Bolt	34 - Oil lever indicator	52 - Converter bypass valve spring
16 - Lock tab	35 - Flat-head screw	53 - Converter bypass valve ball
17 - Bearing retainer	36 - Lock washer	54 - Orifice plug
18 - Bearing	37 - Bolt	
19 - Converter pump hub		

Legend for Figure 10-18

TABLE 10-2
CONVERTER, TORQUE TABLE

<u>Part</u>	<u>Torque Value lb. ft.</u>
Oil cooler cover bolts	15 to 18
Sump plate bolts	15 to 18
Drive element cover nuts	19 to 23
Output shaft bolt	83 to 100
Output shaft bearing retainer bolts	81 to 97
Disconnect clutch yoke bolts	26 to 32
Disconnect clutch lever bolt	67 to 80
Converter housing bolts	26 to 32



- 1 - Bolt
- 2 - Oil cooler cover
- 3 - Oil cooler cover gasket
- 4 - Oil cooler core
- 5 - Oil cooler gasket
- 6 - Orifice plug
- 7 - Coolant drain plug

Figure 10-19. Oil Cooler-Exploded View

3. Secure the cover with twelve bolts (1). Torque bolts as specified in table 10-2.

10-33. SUMP PLATE. (Refer to figure 10-18). The sump plate is removed and reinstalled as follows:

a. To remove sump plate, use a 1/2-inch wrench and remove nine bolts (37) and lock washers (36) from sump plate (38). Remove the sump plate and gasket (39) from housing assembly (44).

10-14

b. Reinstall sump plate as follows:

1. Turn the converter housing on its side and install gasket (39). Install sump plate (38).

2. Using a 1/2-inch wrench, install nine bolts (37) and lockwashers (36) through the sump plate and into the converter housing. Torque the bolts as specified in table 10-2.

10-34. CONVERTER DRIVE ELEMENT (Refer to figure 10-18.) The converter drive element is removed and reinstalled as follows:

a. To remove drive element, use a 1/2-inch wrench and remove the 24 nuts (4) from the gear and cover assembly. Remove the cover (5) and seal ring (3).

b. To reinstall the drive element, place the cover (5) and seal ring (3) on the converter pump assembly (23). Using 1/2-inch wrench, tighten the 24 nuts (4). Torque the nuts as specified in table 10-2.

10-35. CONVERTER ELEMENTS. The converter elements are removed, disassembled, and reinstalled as follows:

a. To remove converter elements, proceed as follows:

1. Remove turbine assembly (7, figure 10-18) and bearing assembly (6).

2. Grasp stator assembly (8) and rotate it counterclockwise to lock it to freewheel roller race (12). Remove the stator and race from the ground sleeve.

NOTE

If the race separates from the stator assembly, rollers (10) and roller springs (11) will drop out of the stator. There are ten rollers and ten roller springs which should be accounted for before continuing with the disassembly.

3. Using snap ring pliers, remove snap ring (13) and roller race spacer (14).

4. Remove the converter pump assembly, items (15) through (23). Also refer to figure 10-20.

b. Disassemble the stator as follows:

1. Remove any springs (11, figure 10-18) or rollers (10) from the stator assembly, which may have remained after removal from the converter.

2. Do not remove thrust bearing (9) unless replacement is necessary.

NOTE

After thorough cleaning, and lubrication with transmission fluid, freewheel roller race (12) can be installed to check for the roughness of bearing (9). Install roller race (12) shouldered side first, and rotate it while pressing it against bearing (9). If roughness is felt, the bearing should be replaced.

3. If bearing (9) must be replaced, remove it carefully to avoid damaging the aluminum bore which holds it.

4. Do not disassemble the stator further. If inspection reveals any

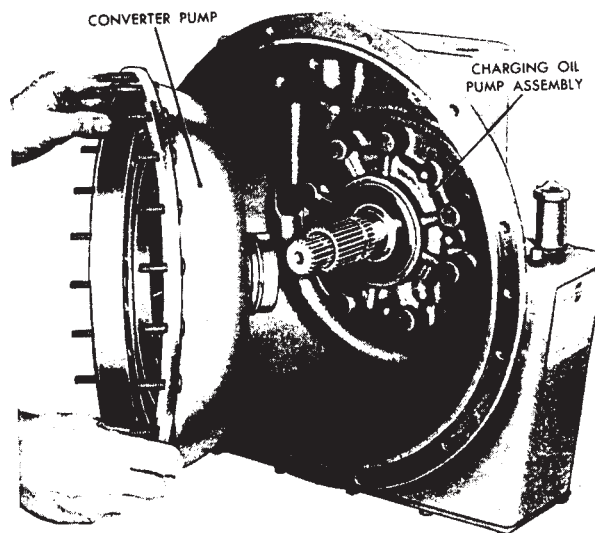


Figure 10-20. Converter Pump Removal

damage or excessive wear, replace the assembly.

c. Assemble the stator as follows:

1. If bearing (9) was removed, install a new bearing assembly. The bearing must be installed as shown in figure 10-21. A special tool should be used to ensure proper assembly (see figure 7-1).

2. Coat the cam pockets in the stator with oil-soluble grease. Install rollers (10, figure 10-18) into the shallow ends of the cam pockets. Install springs (11) into the deep ends of the pockets with one open end toward the inside diameter of the stator (see figure 10-22).

3. Install freewheel roller race (12, figure 10-18) into the stator assembly. Rotate the race clockwise while installing it. This will roll the rollers towards the deep ends of the pockets, allowing the race to enter the stator assembly.

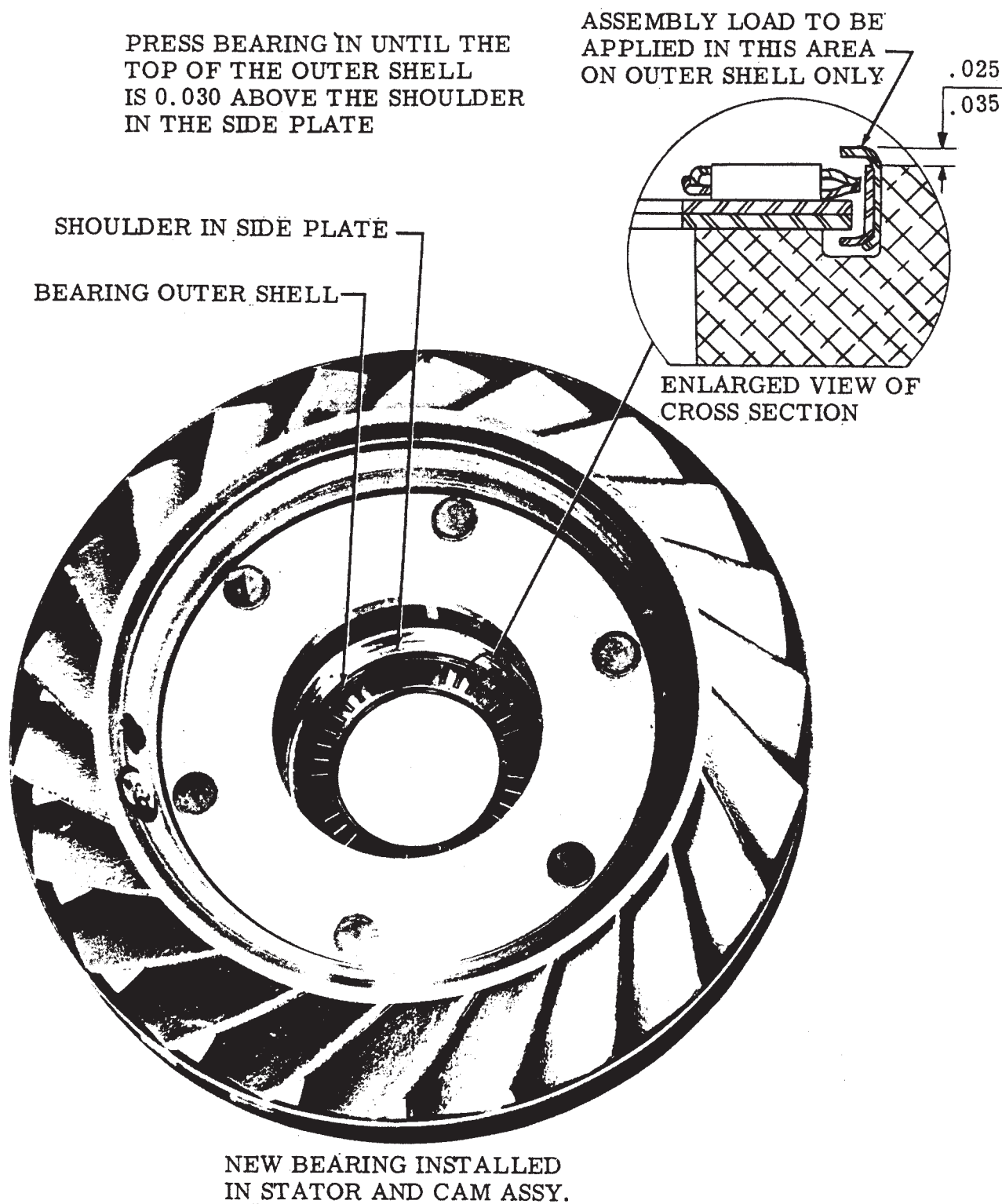


Figure 10-21. Stator Thrust Bearing Installation

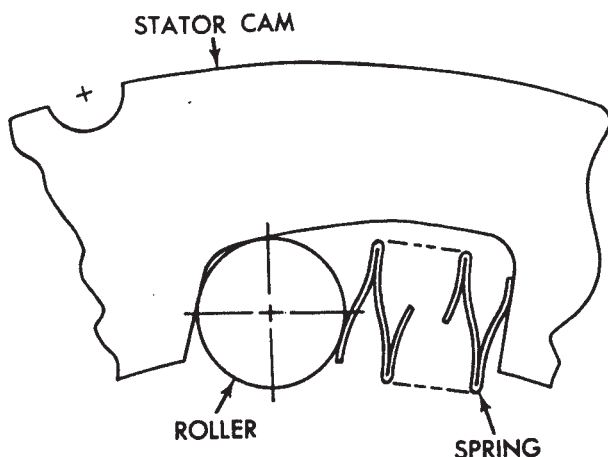


Figure 10-22. Stator Cam Spring and Roller-Rear View

4. When the roller race is firmly seated against the thrust bearing, twist it firmly in a counterclockwise direction to lock it into place. Lay the stator assembly aside until it is ready for installation into the converter.

d. Reinstall the converter elements as follows:

1. Install converter pump assembly (15 thru 24, figure 10-18).
2. Install spacer (14) and snap ring (13) onto the ground sleeve.
3. Install as a unit, stator assembly (8), rollers (10), springs (11) and race (12).
4. Install turbine assembly (7) and bearing (6).

10-36. OUTPUT SHAFT. The output shaft is removed, disassembled and reinstalled as follows:

a. To remove output shaft, take out eight self-locking bolts (7, figure 10-24)

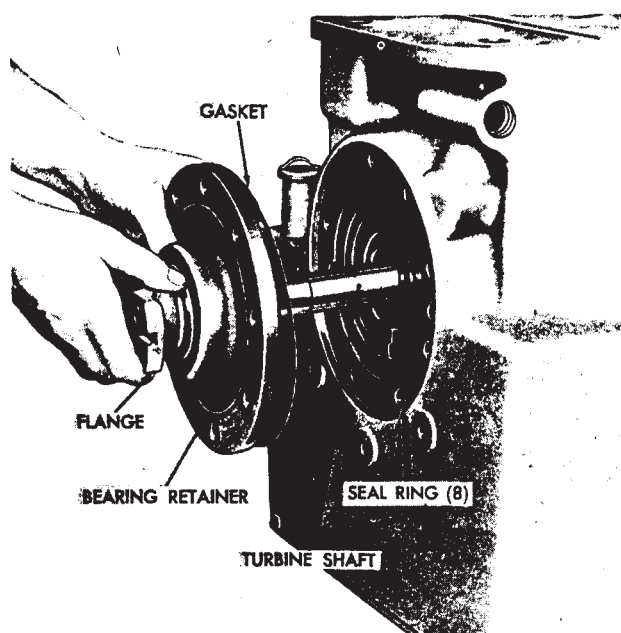


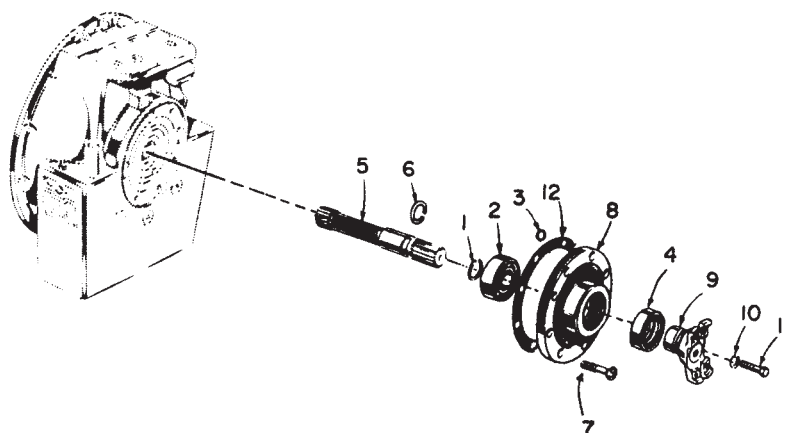
Figure 10-23. Output Shaft Removal

in retainer (8). Remove bearing retainer, gasket, flange, ball bearing assembly and turbine shaft as shown in figure 10-23.

b. Disassemble the output shaft as follows:

1. Using a 3/4-inch wrench, remove bolt (11, figure 10-24) and lock-washer (10) that secure flange (9) to shaft (5).
2. Remove flange (9), bearing retainer (8) and bearing assembly (2) from shaft (5).
3. Remove seal ring (6) and snap ring (1) from shaft (5).
4. Do not remove oil seal (4) from the bearing retainer unless replacement is necessary. If necessary, press the seal from the housing.

c. Assemble the output shaft as follows:



- | | |
|--------------------------|----------------------|
| 1. Snap ring | 7. Self-locking bolt |
| 2. Bearing assembly | 8. Bearing retainer |
| 3. Seal ring | 9. Output flange |
| 4. Oil seal | 10. Lock washer |
| 5. Turbine shaft | 11. Bolt |
| 6. Seal ring (hook type) | 12. Gasket |

Figure 10-24. Output Shaft-Exploded View

1. If oil seal (4, figure 10-24) was removed, press a new seal into bearing retainer (8), with the spring-loaded side next to bearing (2). The rear surface of the seal should be flush with the rear of the retainer.

2. Install snap ring (1) and ball bearing (2) onto shaft (5), positioning the bearing against the snap ring.

3. Install bearing retainer (8) onto bearing (2).

4. Install flange (9) onto shaft (5).

5. Install 1/2-inch bolt (11) and lock washer (10) through flange and into shaft (5). Torque the bolt as specified in table 10-2.

6. Install seal ring (6) onto shaft (5).

d. Reinstall the output shaft in the following manner:

1. Install seal rings, bearing retainer, gasket, flange, ball bearing assembly, and turbine shaft (figure 10-23).

2. Secure the assembly with eight self-locking bolts (7, figure 10-24) through the bearing retainer. Torque the bolts as specified in table 10-2.

10-37. CHARGING OIL PUMP. The charging oil pump is removed, disassembled, and reinstalled as follows:

a. The charging oil pump is removed as follows:

1. Remove two flat-head screws (35, figure 10-18) from the rear of converter housing assembly (42).

2. Thread two of eight bolts (23, figure 10-24) through the converter housing into the pump body. Tap lightly with a soft hammer on the bolt heads until the charging oil pump assembly is loose. Remove the bolts and then remove the charging oil pump assembly. (See figure 10-20.)

b. Disassemble the charging oil pump as follows:

1. Remove seal ring (31, figure 10-18) from the pump body (28).

2. Using an Allen wrench, remove three screws (33) from ground sleeve (32).

3. Remove the converter ground sleeve as shown in figure 10-25.

4. Remove the drive gear and the driven gear (29 and 30) from the pump body.

NOTE

The gears are a matched set and are not serviced separately.

5. If necessary for parts replacement, drive out oil seal (27) from pump body (28).

c. Reassemble the charging oil pump as follows:

1. If oil seal (27, figure 10-18) was removed from body (28) install a new seal, lip side first.

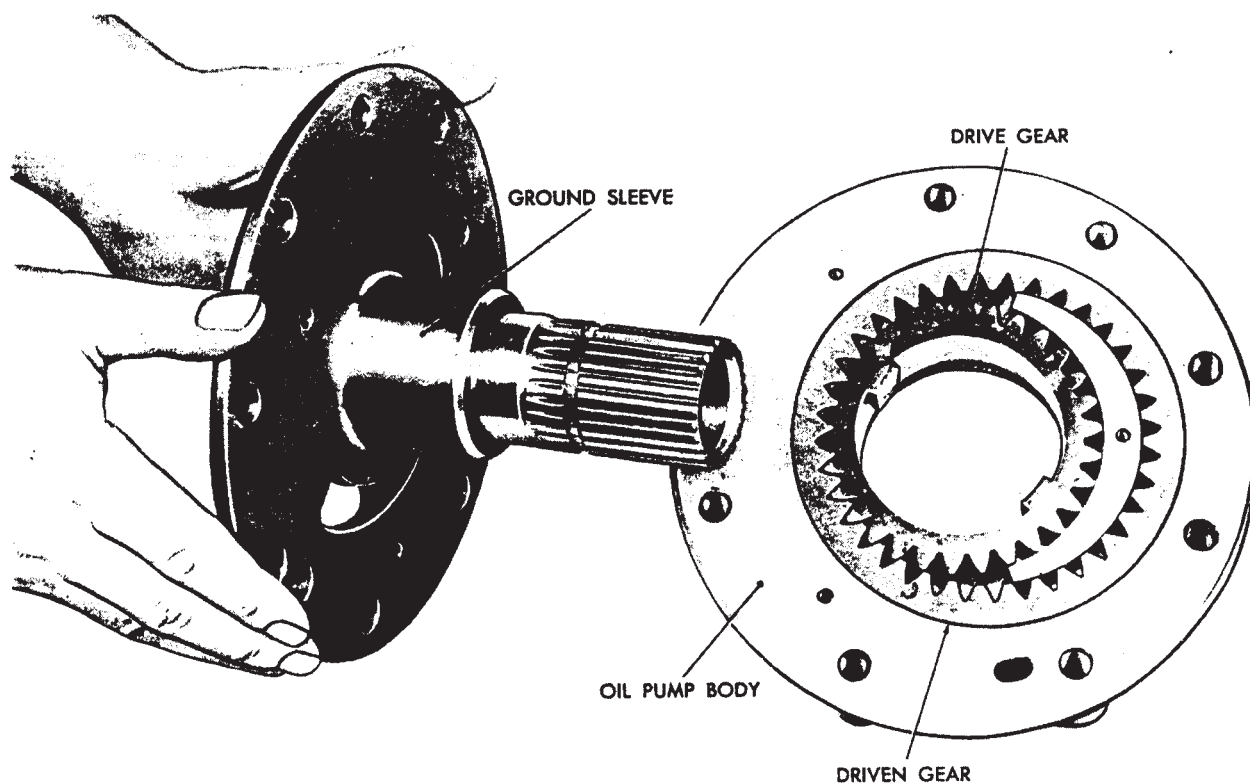


Figure 10-25. Converter Ground Sleeve Removal

2. Install gears (29) and (30) into oil pump body (28).

3. Install converter ground sleeve (32) into oil pump body (28).

4. Install three screws (33) into the rear of ground sleeve (32) to secure the sleeve to the pump body.

5. Install seal ring (31) onto oil pump body (28).

d. Install the charging oil pump as follows:

1. Install two headless guide bolts into the rear of charging oil pump assembly (25).

2. Grease seal ring (31) with heavy grease.

3. Install the charging oil pump assembly into the front of converter housing (42). Secure the charging oil pump with two flat-head screws (35).

10-38. CONVERTER BYPASS VALVE. The converter bypass valve is removed and reinstalled in the following manner:

a. To remove the bypass valve, refer to figure 10-18 and remove plug (51), spring (52), and ball (53) from cooler mounting pad of converter housing.

b. Install the bypass valve as follows:

1. If plug (6, figure 10-19) was removed, press in a new plug.

2. Install ball (53, figure 10-18), spring (52), and plug (51), into converter housing (44).

10-39. INPUT DISCONNECT CLUTCH OVERHAUL.

10-40. REMOVAL. If the torque converter assembly and the input disconnect clutch assembly are removed as a unit, separate the two assemblies by removing the twelve bolts (14, figure 10-27) and lock washers (13) that secure the converter housing to the disconnect clutch housing.

10-41. DISASSEMBLY. The input disconnect clutch is disassembled in the following manner:

a. Disassemble the disconnect clutch components as follows:

1. Use an 11/16-inch wrench to prevent flexible lubrication tube assembly (27, figure 10-26) from turning inside housing; use a 15/16-inch wrench to remove jam nut (26) on the outside of the tube assembly. Push the tube assembly to the inside of the housing. The housing is marked for proper installation of the tube.

2. Place clutch control hand lever (34) in released (rearward) position and remove the 3-segment facing (7).

3. Using a 3/4-inch wrench, remove bolt (36) and lockwasher (35) from hand lever (34). Remove the lever.

4. Place the input disconnect clutch on its rear face (refer to figure 10-28).

5. Straighten the tab on lockwasher (3, figure 10-26).

6. Using a 2 3/8-inch wrench, remove nut (2) and lockwasher (3).

7. Remove the clutch assembly and the key from the shaft (refer to figure 10-28).

8. Using a 9/16-inch wrench, remove two bolts (29, figure 10-26) and lockwashers (30) from the clutch shifter yoke (31).

9. Remove the clutch shifter shaft (33), from either side. Two Woodruff keys (32) can be removed from the shaft as the shaft is removed from the yoke and housing (also see figure 10-29).

10. Using a 1/2-inch wrench, remove tube assembly (27) from release bearing (22).

11. Remove the external snap ring (shown in figure 10-29) from the shaft.

12. Place the housing in a vertical position.

13. Remove shaft assembly (see figure 10-29).

14. Do not remove ball bearing assembly (3, figure 10-27) from housing (4) unless replacement is necessary. When necessary remove internal snap ring (2) and remove the bearing. Install the new bearing, shielded side toward snap ring (2), and install the snap ring.

15. Do not remove drive ring (12) from the shaft assembly (8) unless replacement is necessary. When necessary, remove eight 3/8-inch bolts (11), nuts (9) and washers (10), using a 9/16-inch wrench, and remove input-drive ring (12).

b. Disassemble the clutch assembly (removed in step 7 above) as follows:

1. Do not disassemble clutch assembly (figure 10-26) unless replacement of parts is necessary. When necessary, remove two bolts (24) and nuts (23) from release bearing assembly (21) and remove bearing (22).

2. Remove lever spring (14).

3. Remove retainer rings (15) from the three clevis pins (13) that secure the levers (16) to clutch body (5).

4. Remove the three pins (13) and remove release sleeve (20) and levers (16).

5. Remove the remaining two retainer rings (18) from each of the three levers (16) and remove the levers, clevis pins (19), and connecting links (17).

6. Remove the separator springs (6) and pressure plate (8).

7. Remove fillister-head screw (11), shakeproof washer (10), and adjusting lock (9).

8. Thread adjusting nut (12) out of pressure plate (8).

10-42. REASSEMBLY. Reassemble the clutch assembly and the input disconnect clutch in the following manner:

a. The clutch assembly (figure 10-26) is reassembled as follows:

1. Thread adjusting nut (12) into the pressure plate (8).

2. Lay clutch body (5) on the table with the clutch facing side up.

3. Install the three separator springs (6) into clutch body (5).

4. Install pressure plate (8) on separator springs (6).

5. Install the three levers (16) and three clevis pins (13) on clutch body (5) and install the three retainer rings (15) on clevis pins (13).

6. Install the six connecting links (17) and the three clevis pins (19) on levers (16). Install the three retainer rings (18) on clevis pins (19).

7. Install release sleeve (20) and three clevis pins (19) to the connecting links (17). Install three retainer rings (18) on clevis pins (19).

8. Install lever spring (14).

9. Install release bearing (22) and secure it with two 5/16-inch bolts (24) and nuts (23).

10. Turn adjusting nut (12) to a position that will allow the cam part of levers (16) to be past center when the clutch is in the applied position and the 3-segment facing is installed. This facing must be held tightly in the clutch.

NOTE

Position the facing segments so that, as the clutch is applied, the

inner diameter edge of the segments will not be broken by the shoulder on the clutch body.

11. Do not install the lock on the adjusting nut until after final adjustment has been made in b, steps 13 thru 16, below.

b. Reassemble the input disconnect clutch as follows:

1. If drive ring (12, figure 10-27) was removed, install it. Retain it with eight bolts (11), washers (10) and nuts (9). Install shaft assembly (8) into bearing (3). Bearing was installed in paragraph 10-41, step b, 14.

2. Install snap ring (1) into the groove in shaft assembly (8).

3. Install tube assembly (27, figure 10-26) into release bearing (22).

1 - Drive ring
2 - Nut
3 - Washer
4 - Manual input disconnect clutch assembly
5 - Clutch body
6 - Separator spring
7 - Clutch facing assembly
8 - Pressure plate
9 - Adjusting lock
10 - Lock washer
11 - Screw
12 - Adjusting nut
13 - Clevis pin
14 - Lever spring (set)
15 - Retainer ring
16 - Lever
17 - Connecting link
18 - Retainer ring

19 - Clevis pin
20 - Release sleeve
21 - Release bearing assembly
22 - Release bearing
23 - Stop nut, 5/16-24
24 - Bolt, 5/16-24 x 1 3/4
25 - Grease fitting
26 - Jam nut
27 - Flexible lubrication tube assembly
28 - Key
29 - Bolt
30 - Lock washer
31 - Clutch shifter yoke
32 - Woodruff key
33 - Clutch shifter shaft
34 - Clutch control lever
35 - Lock washer
36 - Bolt

Legend for Figure 10-26

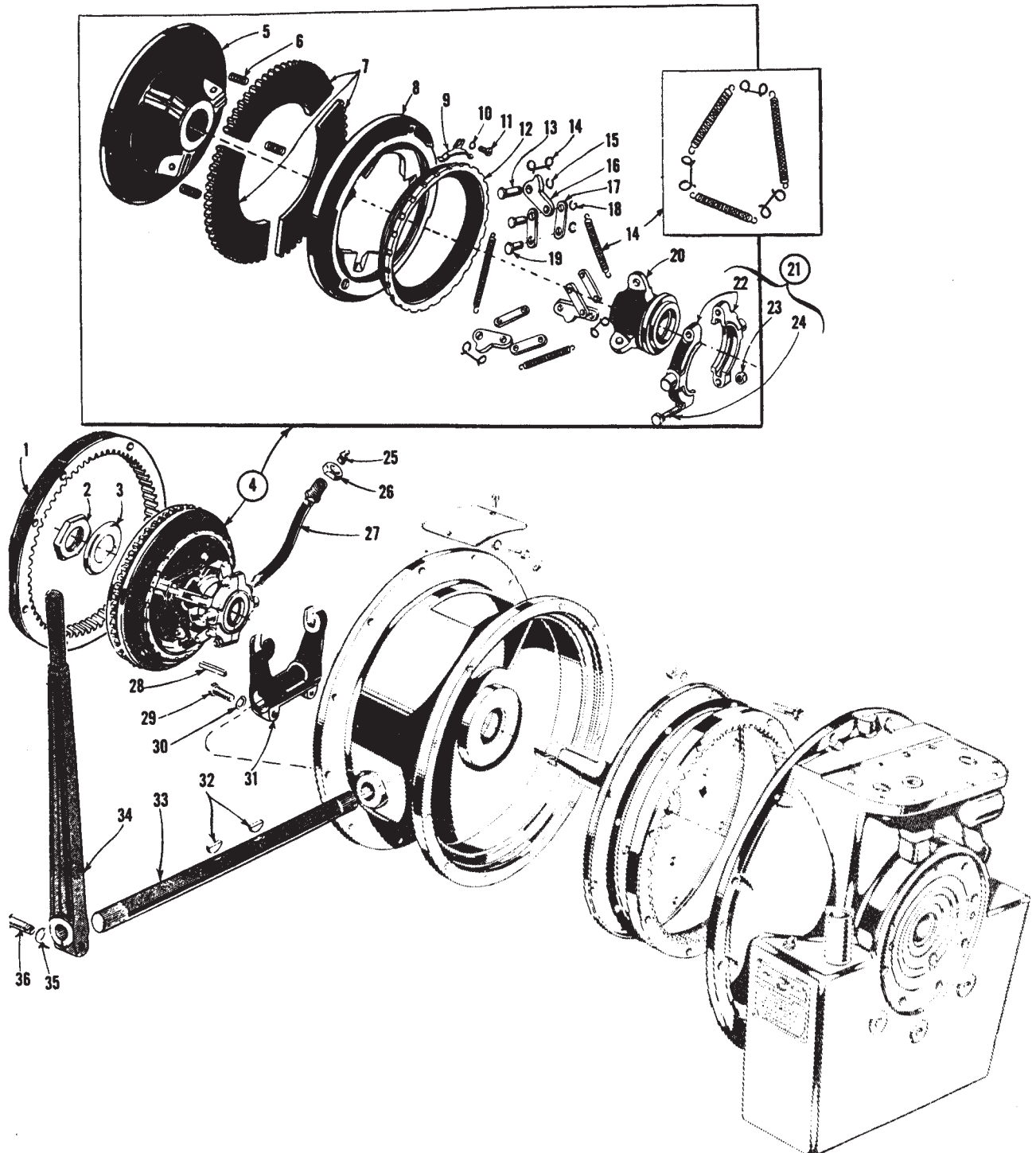
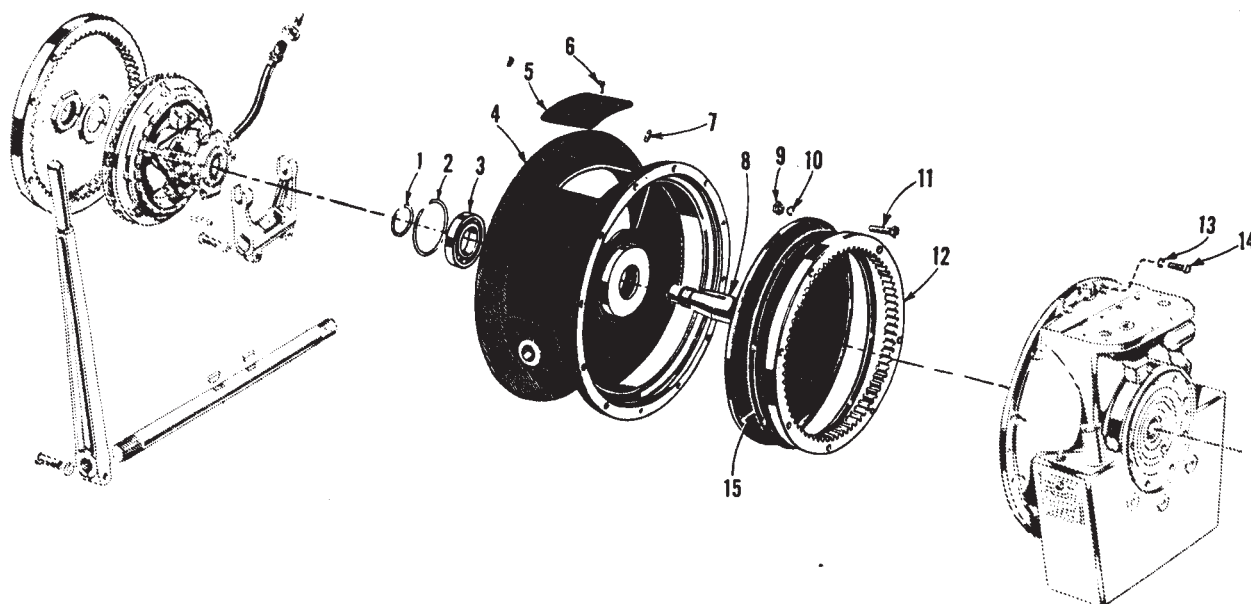


Figure 10-26. Input Disconnect Clutch-Exploded View



- 1 - Snap ring
- 2 - Snap ring
- 3 - Bearing assembly
- 4 - Manual input disconnect clutch housing
- 5 - Housing cover
- 6 - Screw
- 7 - Grease fitting
- 8 - Converter input shaft assembly
- 9 - Nut
- 10 - Lock washer
- 11 - Bolt
- 12 - Drive ring
- 13 - Lock washer
- 14 - Bolt
- 15 - Gasket

Figure 10-27. Input Disconnect Clutch Housing and Drive Ring-Exploded View

4. Install clutch shifter shaft through one bore in the housing. Install two Woodruff keys and the clutch shifter yoke on the shaft, and then push the shaft through the outer bore in the housing (figure 10-27).

5. Center shaft (33, figure 10-26) in the housing and center yoke (31) on the shaft. Install two 3/8-inch bolts (29),

with washers (30), in the yoke and torque them as specified in table 10-2.

6. Use a good quality heavy grease on key (28) and install the key in the clutch assembly keyway.

7. Install the clutch assembly and the key on the shaft assembly. The

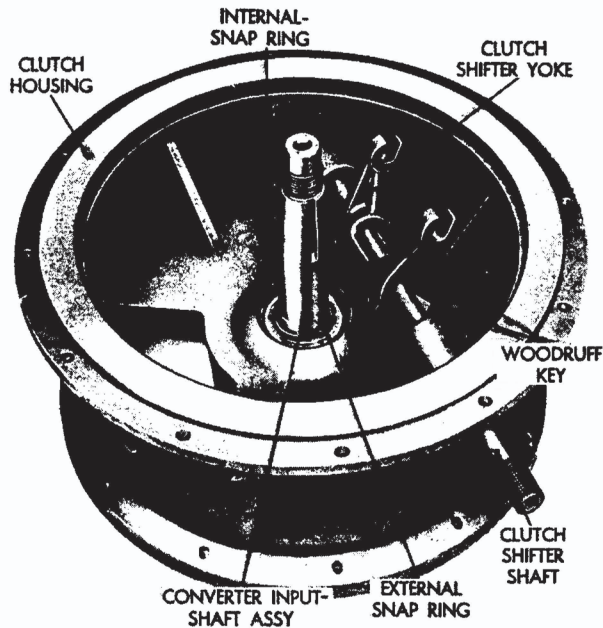


Figure 10-28. Clutch Housing Removal

yoke must be engaged with the clutch release bearing as the clutch assembly is lowered onto shaft (figure 10-28).

8. Install lockwasher (3, figure 10-26) and nut (2) on the end of shaft assembly (8, figure 10-27). Position the washer with the tab in the keyway. Tighten the nut.

9. Using a drift, bend a section of the washer against a flat on the nut.

10. Install clutch control hand lever (34, figure 10-26).

11. Install 1/2-inch bolt (36) and washer (35) into lever (34) and torque the bolt as specified in table 10-2.

12. Push lubrication tube (27) through the hole in the housing and install jam nut (26) on the tube. Prevent the tube assembly from turning by holding the tube fitting inside the housing with an 11/16-inch wrench. Tighten the jam nut with a 15/16-inch wrench.

13. Install 3-segment facing (7) in clutch (4) and check the effort required to move lever (34) into or out of engagement. A suggested method of making this check is to use a hand scale. Put the hook of the scale over the shoulder at the hand grip of the lever. At this point on the lever, 40 to 44 pounds of effort should move the lever into engagement.

14. If the effort required to move the lever is less than 40 pounds, increase the effort by turning the adjusting nut counterclockwise with a spanner wrench or drift and hammer.

15. If the effort required to move the lever is more than 44 pounds, reduce the effort by turning the adjustment nut clockwise.

16. Install the lock, washer, and screw in the front plate (figure 8-12).

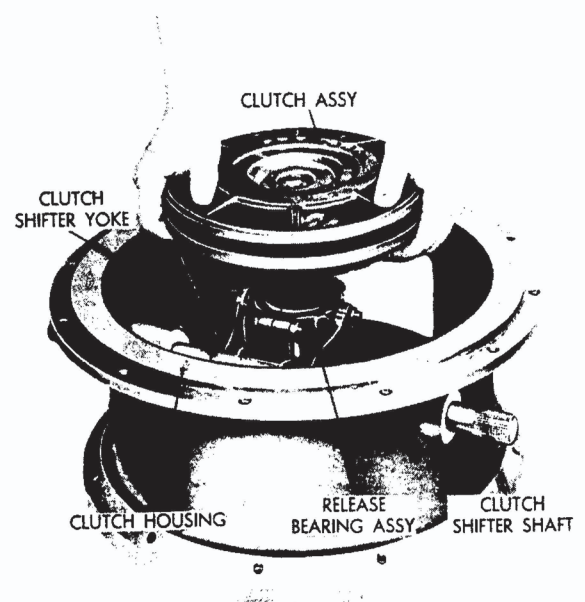


Figure 10-29. Clutch Shifter Shaft and Yoke Installation

DIMENSIONAL POINTS

WEAR LIMIT

B TO EXCEED A	NO LESS THAN 0.005"
D TO EXCEED C	NO LESS THAN 0.005"
*E TO EXCEED F	NO LESS THAN 0.005"
H (ROLLER RACE)	TO PROJECT BEYOND SUR- FACE G (ROTOR SIDE PLATE) 0.007" MINIMUM
G (CLEARANCE BETWEEN THRUST WASHER AND ROTOR SIDE PLATE)	0.007" MINIMUM

*INSERT FEELER GAGE BETWEEN SHROUD DIAMETER OF
STATORS

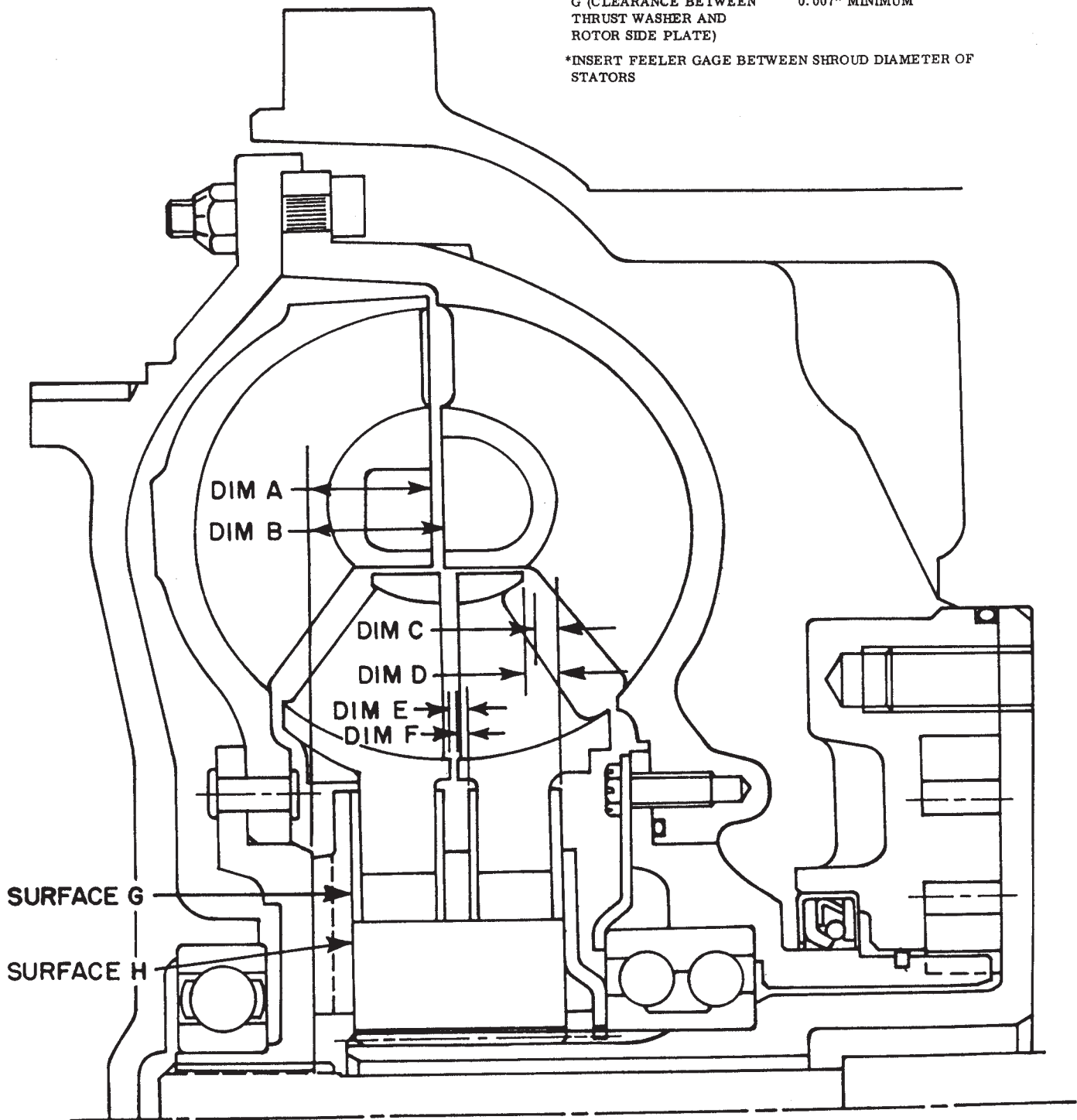


Figure 10-30. Torque Converter Wear Limits Points of Measurements

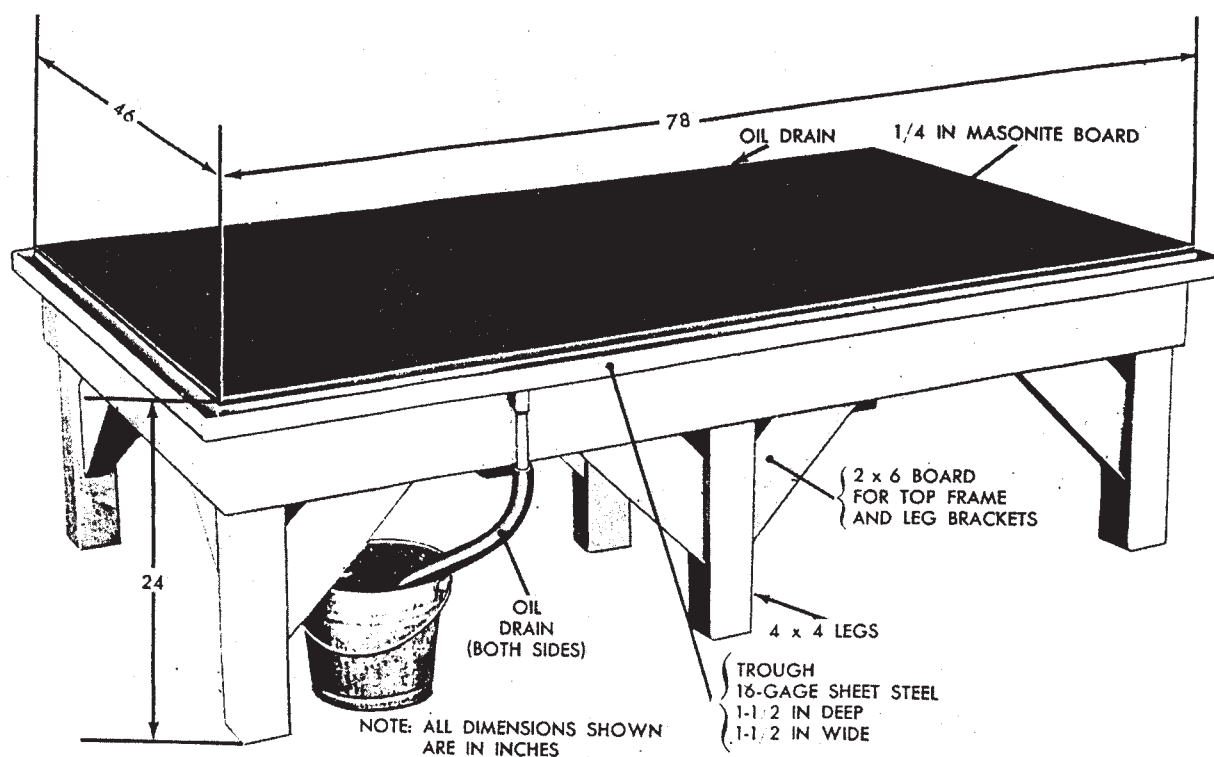


Figure 10-31. Torque Converter Disassembly Table

10-43. INSTALLATION. If the torque converter assembly and the input disconnect clutch assembly were removed as a unit, assemble the two units by placing the rear face of the input disconnect clutch housing against the front face of the con-

verter housing. Make sure the gear and cover assembly on the converter is aligned with input drive ring (12, figure 10-27). Install twelve 3/8-inch bolts (14) and washers (13) through the converter housing. Torque the bolts as specified in table 10-2.

[illegible]

Figure 10-32. Torque Converter Overhaul Stand

TABLE 10-3
CONVERTER, GENERAL BOLT AND SCREW
TORQUE SPECIFICATIONS

Size	Threads per inch	Standard heat-treated bolts and screws	Special heat-treated bolts, screws, Allen-head screws, and self-locking capscrews
			  
1/4	20	6-8	9-11
	28	8-10	10-12
5/16	18	15-18	17-20
	24	17-20	19-23
3/8	16	26-32	36-43
	24	33-40	41-49
7/16	14	42-50	54-65
	20	50-60	64-77
1/2	13	67-80	81-97
	20	83-100	96-115
9/16	12	85-100	103-123
	18	100-120	122-146
5/8	11	117-140	164-192
	18	134-160	193-225
3/4	10	180-210	284-325
	16	215-250	337-385
7/8	9	315-360	490-550
	14	372-425	575-650
1	8	445-500	685-770
	14	535-600	830-925

NOTE

All torque values are
given in lb. ft.

TABLE 10-4

CONVERTER OVERHAUL INSPECTION

Part	Inspect for	Action To Be Taken
Machined surfaces of cast parts	Wear, scratches, grooves, and dirt.	Remove scratches and burs with crocus cloth and remove foreign matter. Replace deeply grooved or scratched parts.
	Oil passages for obstruction.	Remove obstruction with compressed air or by working a wire back and forth through passage and flushing out with cleaning solvent.
	Mounting faces for nicks, burs, scratches, and foreign matter.	Remove defects with crocus cloth or soft stone. Replace part if scratches are deep.
	Damaged threaded openings.	Chase damaged threads with correct size tap.
	Cracks in housings or other cast parts.	Replace cracked parts.
	Machined surfaces for any damage that could cause oil leakage or malfunction.	Rework or replace defective parts.
Bearings	Roughness of rotation.	Replace bearing if rotation still rough after cleaning and oiling.
	Scored, pitted, scratched, cracked, or chipped races and excessive roller or ball wear.	Replace bearing if any such defect found.
	Grooved, burred, or galled conditions on housing and shaft of any defective bearing (indication that bearing has been turning).	If damage cannot be repaired with crocus cloth, replace defective part.

TABLE 10-4

CONVERTER OVERHAUL INSPECTION (Cont.)

Part	Inspect for	Action To Be Taken
Gears	Scuffed, nicked, burred, or broken teeth. Wear that destroys original tooth shape. Scores, scratches and burs on thrust faces of gears.	Replace gear if defect cannot be repaired with soft stone. Replace defective gear. Replace gear if defect cannot be removed with soft stone.
Splined parts	Stripped, twisted, chipped, or burred splines. NOTE Spline wear is not considered detrimental unless tightness of fit on splined parts is affected.	Remove burs with soft stone. Replace part if other defects are found.
Threaded parts	Burred or damaged threads.	Remove burs with soft stone or fine file. Replace damaged parts.
Snap rings	Nicks, distortion, and excessive wear. NOTE Snap ring must snap tight in its groove for proper functioning.	Replace defective snap ring.
Springs	Signs of overheating, permanent set, or wear due to rubbing adjacent parts.	Replace defective spring.

TABLE 10-4

CONVERTER OVERHAUL INSPECTION (Cont.)

Part	Inspect for	Action To Be Taken
Bushings	Scores, sharp edges, roundness, burs and evidence of overheating. NOTE It may sometimes be necessary to cut out a defective bushing. Care must be taken not to damage the bore that holds bushing.	Remove scores with crocus cloth. Remove burs and sharp edges with a scraper or knife blade. Replace out-of-round, deeply scored, or worn bushing.
Thrust washers	Distortion, scores, burs and wear.	Replace defective thrust washer.
Seal rings	Cuts and hardness. CAUTION When replacing lip-type seal rings, make sure spring-loaded side is toward oil to be sealed in (toward inside of unit). Use non-hardening sealing compound on the outside diameter of seal to aid in prevention of oil leaks. Wear, broken hooks, and distortion on hook-type seal rings. NOTE Hook-type seal rings are worn excessively if no gap exists between the hooks on installation.	Replace defective seal rings.
Composition gaskets		Replace all composition gaskets.

TABLE 10-5
TORQUE CONVERTER SPRING WEAR LIMITS DATA

Spring	Part Number	Data
Freewheel roller (11, figure 10-18)	6774966	6-1/2 coils, corrugated type 0.0024-0.0026 wire diam. 0.896 width 0.64 free length 0.30 length under 0.15 lbs. load
Converter bypass valve (52, figure 10-18)	6758597	13 coils 0.118-0.122 wire diam. 0.794 outside diam. 2.81 free length 2.11 length under 57.2 to 63.6 lbs. load
Clutch separator (6, figure 10-26)	5193431	0.437 outside diam. 1.63 free length 0.812 length under 15 to 20 lbs. load

NOTE

All dimensions are in inches.

